Using MITRE CAR (Cyber Analytics Repository)

Week 7 -MITRE CAR: Implementing CAR-2014-11-004 (Remote PowerShell Sessions) in Splunk

CAR-2014-11-004 detects the creation of a remote PowerShell session by looking for the WinRM session host process, wsmprovhost.exe, spawned by svchost.exe. On Windows, any inbound WinRM-based PowerShell remoting session (Invoke-Command, Enter-PSSession, or third-party tools such as Evil-WinRM) causes the WinRM service to host the session inside a dedicated wsmprovhost.exe process, created as a child of the WinRM service process (svchost.exe). The analytic's data model is a process-creation event with two fields: the process image name and its parent image name.

Per the published analytic, its reported coverage is Moderate for both of the techniques it maps to (see Section 4), and it is listed as related to the D3FEND countermeasure D3-PLA (Process Lineage Analysis) -i.e., detection by examining parent/child process relationships rather than a single process in isolation, which matches exactly what is implemented below.

Implementation in Splunk

1. **Detecting query**

   The analytic's logic (process = wsmprovhost.exe, spawned by svchost.exe) was translated into the following SPL query against Sysmon Event ID 1 (Process Creation) data, as configured in Week 5:

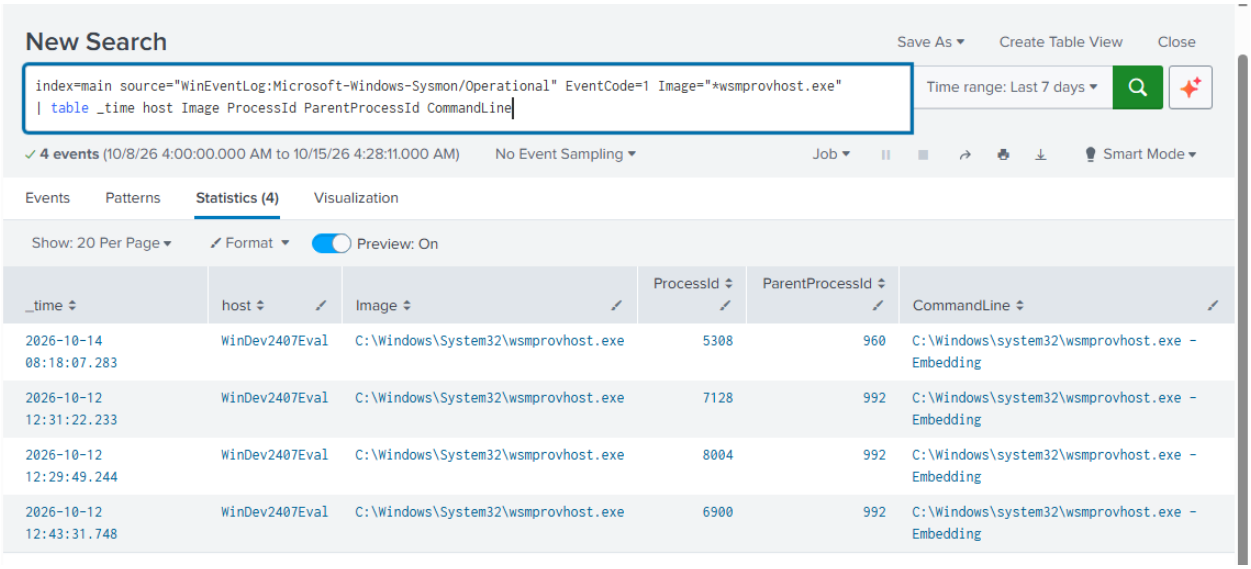


Figure 1. Detecting query results in Splunk: four wsmprovhost.exe sessions found over the last 7 days, including the Week 6 Evil-WinRM session.

2. **Control (negative) query**

To confirm the analytic does not trigger on ordinary local administrative activity, the Week 5/6 control run (whoami, systeminfo, tasklist, net user executed locally, without WinRM) was queried separately:

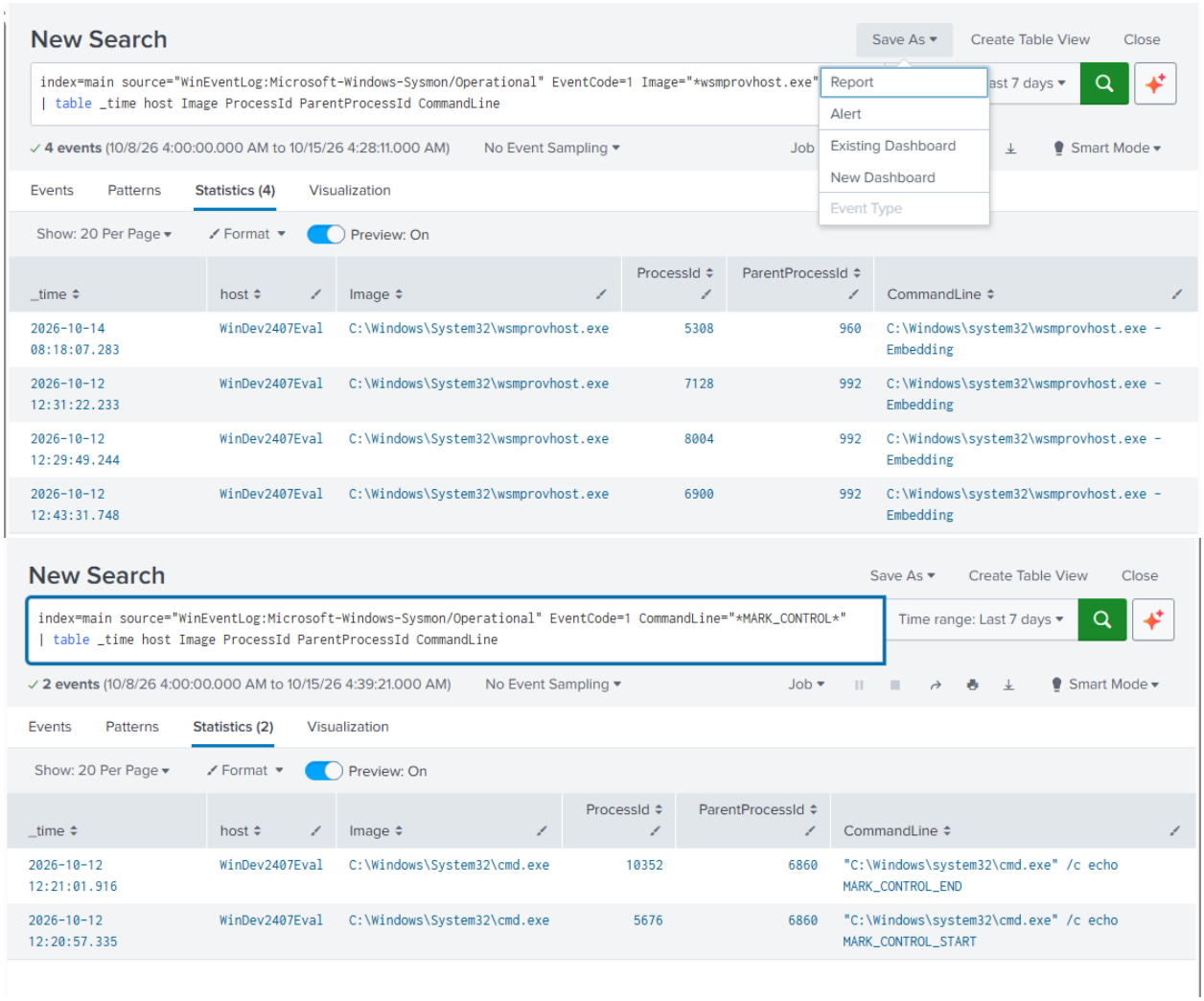
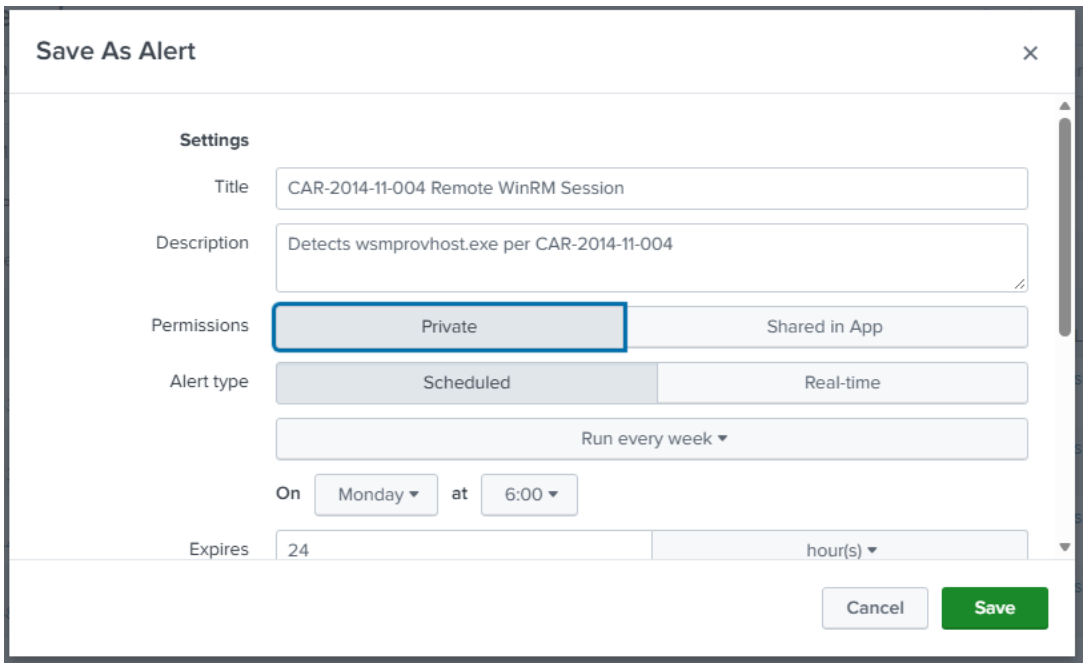


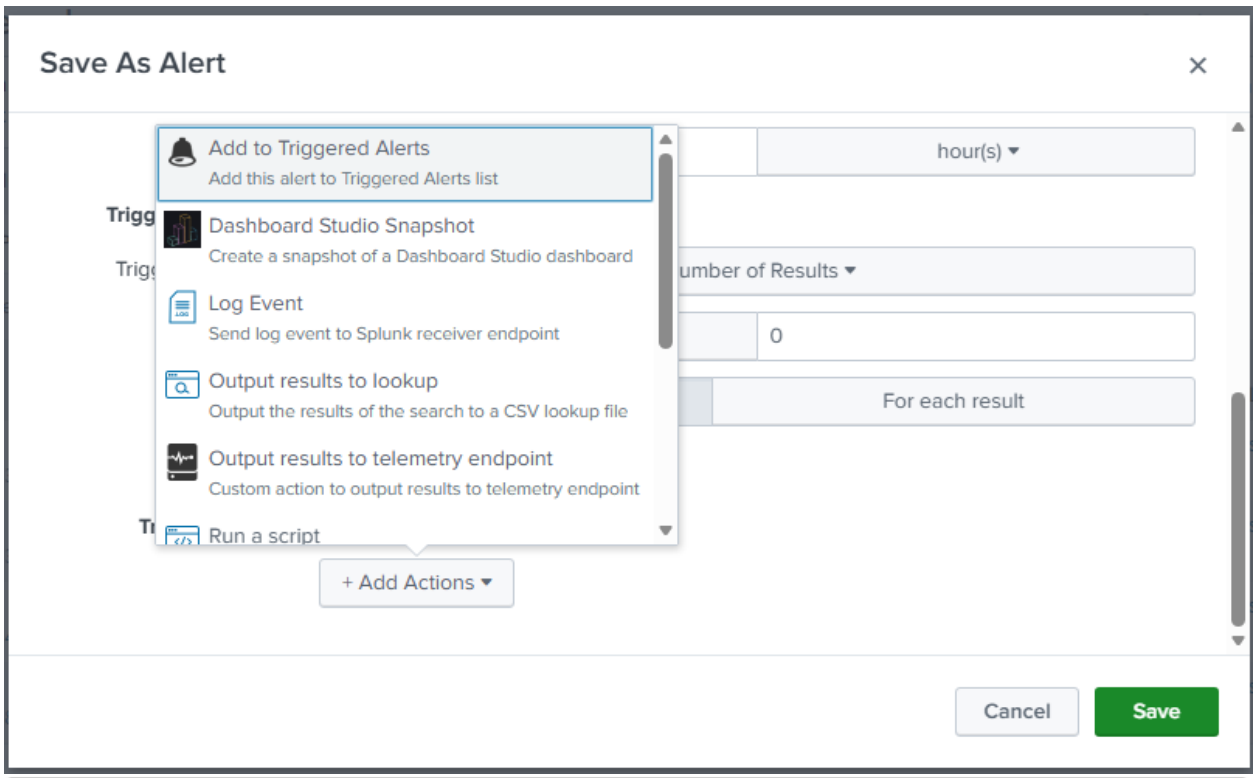
Figure 2. Control query results: the local, non-WinRM control run does not produce a wsmprovhost.exe process and is not flagged by the analytic.

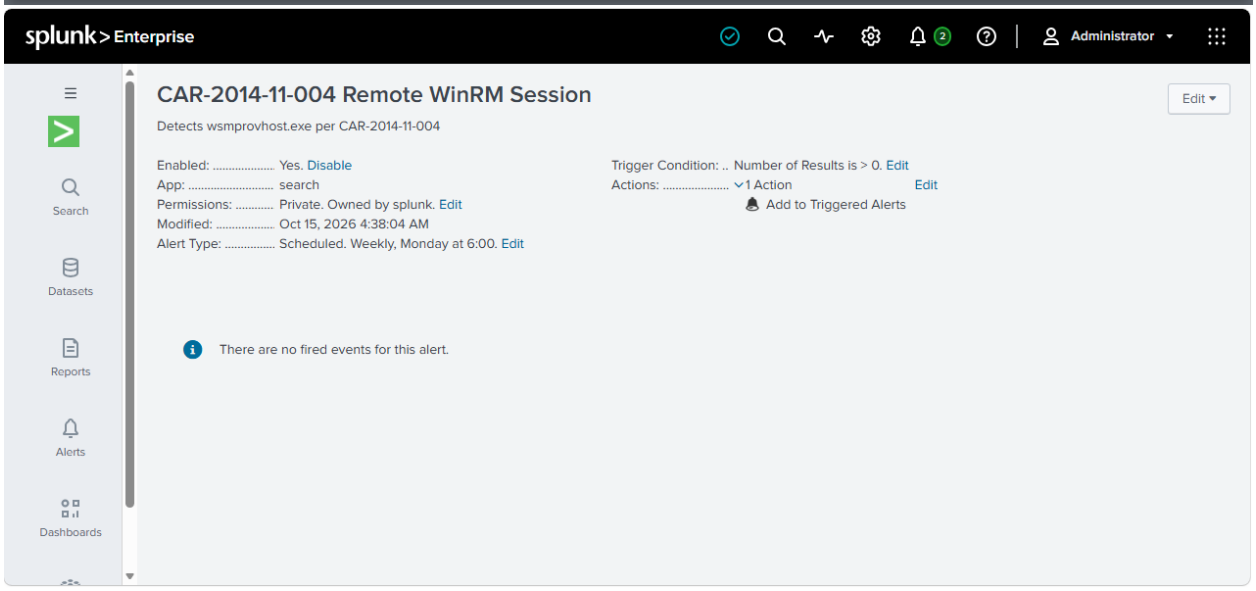
3. Saved as a Splunk Alert

The detecting query (Section 3.1) was saved as a scheduled Splunk Alert to operationalize the analytic:

- Title: CAR-2014-11-004 Remote WinRM Session
- Trigger condition: Number of Results is greater than 0
- Alert type: Scheduled, weekly
- Trigger action: Add to Triggered Alerts (severity: Medium)


Save As Alert
Settings
Title
CAR-2014-11-004 Remote WinRM Session
Description
Detects wsmprovhost.exe per CAR-2014-11-004
Permissions
Private
Shared in App
Alert type
Scheduled
Real-time
Run every week
On
Monday
at
6:00
Expires
24
hour(s)
Cancel
Save





Mapping Analytic to ATT&CK

Analytic	Technique	Tactic
CAR-2014-11-004 (wsmprovhost.exe process creation)	T1021.006 -Remote Services: Windows Remote Management	Lateral Movement
CAR-2014-11-004 (same analytic, session itself is a	T1059.001 -Command and Scripting Interpreter:	Execution

PowerShell execution context)	PowerShell	

This mapping is consistent with the technique chain established in Weeks 2 and 6: T1021.006 covers the remote-access vector, while T1059.001 covers the command execution that occurs inside the resulting session (confirmed in Week 6 via whoami.exe, net.exe, and ipconfig.exe as children of wsmprovhost.exe).

Conclusion

CAR-2014-11-004 was implemented in Splunk against the Sysmon telemetry collected across Weeks 5 and 6, successfully detecting all four WinRM sessions generated in this lab -including the Week 6 dual-VM Evil-WinRM session emulating Storm-0501's lateral-movement behavior - while correctly not triggering on the non-WinRM control run. The detecting query was operationalized as a scheduled Splunk Alert, and the analytic was mapped to T1021.006 (Lateral Movement) and T1059.001 (Execution), consistent with the technique chain built across Weeks 2, 4, and 6. The main limitation carried forward is that the analytic detects the presence of a remote PowerShell session but cannot by itself distinguish a legitimate administrative session from an attacker's, reinforcing the Week 6 finding that WinRM-based lateral movement requires contextual correlation -host identity, source IP, and command content -rather than process telemetry alone.